

**Indian Institute of Information Technology,
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Department of Electronics & Communication Engineering

Smart Attendance System

Using Face Recognition

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0. Abstract

This project presents a Smart Attendance System that uses face recognition to automatically mark student attendance from recorded classroom sessions. The system detects faces using OpenCV, generates face embeddings using the ArcFace model, and matches them using cosine similarity. Attendance is saved in Excel format and emailed to the faculty.

1. Problem Statement

Traditional attendance methods are slow, error-prone, and easy to forge. Paper registers are hard to maintain and analyse. This system eliminates manual roll-call and prevents proxy attendance using biometric face verification.

2. Objectives

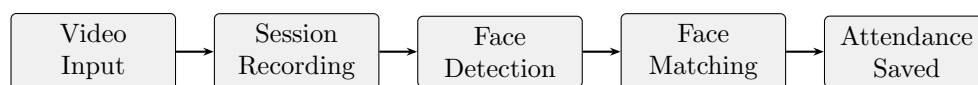
- Automate attendance using face recognition.
- Prevent proxy attendance through biometric verification.
- Store attendance records in Excel format.
- Support webcam, mobile camera, and video file as inputs.
- Automatically email attendance reports to faculty.

3. System Overview

The system has three main modules:

1. **build_db.py** – Builds a face embedding database from student videos.
2. **run_recognition.py** – Records the class session and recognises faces.
3. **send_email.py** – Emails the attendance report to faculty.

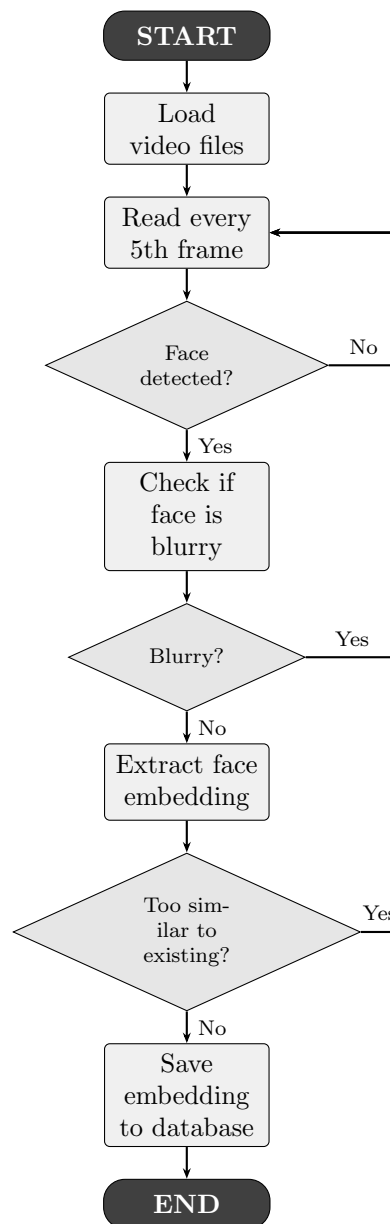
3.0 System Pipeline



After attendance is saved, the report is automatically sent to the faculty via email.

4. Module 1: Building the Face Database

Student video clips (named by roll number) are placed in `data/videos/`. The system processes them to extract and store face embeddings in `face_db.pkl`.

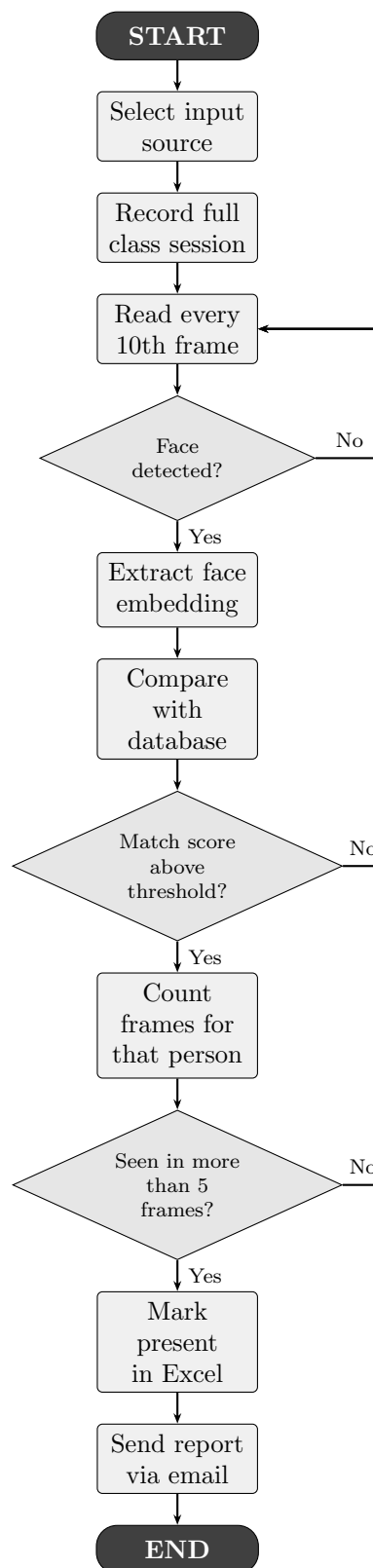


Key behaviours:

- Blurry face images are automatically skipped.
- Only unique (diverse) face views are stored – no near-duplicates.
- New students can be added without deleting existing records.
- A centroid (average embedding) is also stored per person.

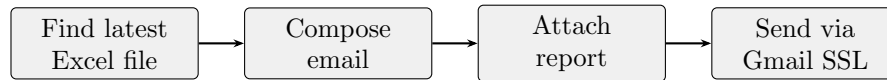
5. Module 2: Face Recognition & Attendance

The full class session is recorded first, then processed to mark attendance. This gives more reliable results than processing live frame-by-frame.



6. Module 3: Email Reporting

After the session, the system automatically picks the latest attendance file and emails it to the faculty using Gmail SMTP.



Note: Gmail 2-Step Verification and an App Password are required. The normal Gmail password will not work.

7. Technology Stack

Tool	Purpose
Python 3	Core programming language
OpenCV	Video capture and face detection
DeepFace (ArcFace)	Face embedding generation
NumPy	Cosine similarity calculations
Pandas	Excel-based attendance records
smtplib	Sending email reports

8. Conclusion

The Smart Attendance System successfully automates classroom attendance using face recognition. The session-based approach, blur filtering, and diversity checks make it accurate and reliable. Attendance is saved and emailed automatically, reducing manual effort for faculty. The system is modular and can be improved further with a GUI and better detection models.